ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

TECHNOLOGY

PAPER 1 · PRACTICE B

MARKS: 75 **TIME: 1½ hours**

Covers the Grade 7 year: design briefs and specifications, bridges, struts, ties and stability, materials and joining methods, levers and linkages, and electrical circuits, energy conversions and magnets.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Use a pencil and ruler for drawings. Drawings must be neat and labelled.
4. Show ALL calculations. A non-programmable calculator may be used.
5. Diagrams are NOT drawn to scale.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 A spider's web is an example of a …

A shell structure. B solid structure. C frame structure. D mass structure.

1.2 A pulling force that stretches a material is called …

A tension. B compression. C torsion. D bending.

1.3 A bridge with a curved shape that carries the load out to its ends is …

A a beam bridge. B an arch bridge. C a suspension bridge. D a cantilever.

1.4 In which class of lever is the effort between the fulcrum and the load?

A First B Second C None D Third

1.5 A machine with a mechanical advantage greater than 1 …

A needs less effort than the load. B needs more effort than the load.
C needs no effort. D cannot move a load.

1.6 Which of these materials is an electrical insulator?

A Aluminium B Silver C Glass D Copper

1.7 Which component changes electrical energy into sound?

A Bulb B Motor C Buzzer D Cell

1.8 A drawing that shows the front of an object flat with the depth lines drawn at 45° is …

A an oblique drawing. B an isometric drawing.
C a perspective drawing. D a plan.

1.9 The person for whom a product is designed is called the …

A designer. B manufacturer. C supplier. D client.

1.10 A list of the requirements that a product must meet is called the …

A design brief. B specifications.
C evaluation. D sketch.

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QUESTION 2: STRUCTURES AND DESIGN

2.1 Name the type of bridge described: (a) a flat deck resting on pillars; (b) a deck that hangs from cables attached to tall towers; (c) a curved bridge that pushes its load out to the ground at each end. (3)

2.2 (a) Explain the difference between a strut and a tie. (b) Give ONE example of a tie in a structure. (3)

2.3 (a) Give TWO ways to make a structure more stable. (b) Why does a tower crane have heavy concrete blocks on the short arm at the back? (4)

2.4 What is a gusset, and why is it used? (2)

2.5 The local municipality wants a shelter for people waiting at a taxi rank.

2.5.1 Write a short design brief for the shelter. (2)

2.5.2 Give TWO specifications and TWO constraints for the shelter. (4)

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QUESTION 3: MATERIALS AND PROCESSING

3.1 Classify each material as natural or manufactured (synthetic): (a) wood; (b) plastic; (c) cotton; (d) steel. (4)

3.2 (a) Give TWO properties of plastic that make it useful for packaging. (b) Give TWO disadvantages of plastic packaging. (4)

3.3 Name ONE method to join: (a) two pieces of wood; (b) two pieces of metal. (2)

3.4 Give TWO safety rules when using a hacksaw. (2)

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QUESTION 4: MECHANICAL SYSTEMS

4.1 Name the class of lever for each tool: (a) a pair of scissors; (b) a nutcracker; (c) a broom while sweeping. (3)

4.2 A wheelbarrow has a load arm of 0,4 m and an effort arm of 1,6 m. (a) Calculate the mechanical advantage. Use: MA = effort arm ÷ load arm. (b) Calculate the effort needed to lift a load of 1 000 N of bricks. (4)

4.3 Third-class levers have a mechanical advantage of less than 1. Explain why they are still useful, and give ONE example. (3)

4.4 (a) Name the linkage in which the input and output move in the same direction. (b) Name ONE use of a bell crank. (2)

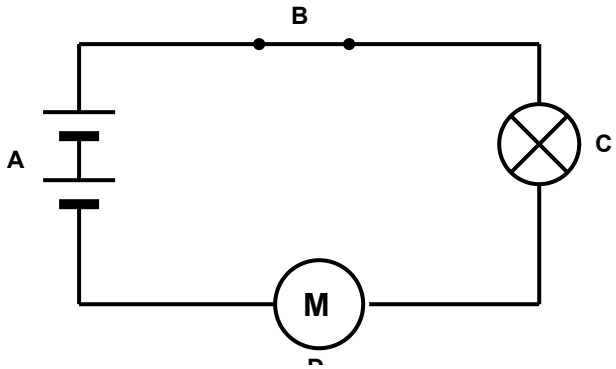
4.5 Draw a systems diagram (input → process → output) for a pair of pliers used to bend wire. (3)

4.6 Draw a neat, labelled sketch of a first-class lever. Show the effort (E), load (L) and fulcrum (F). (3)

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QUESTION 5: ELECTRICAL SYSTEMS

5.1 Study the circuit diagram.



5.1.1 Name the components labelled A, B, C and D. (4)

5.1.2 What happens to C and D when the circuit is as shown? (1)

5.1.3 If the bulb (C) blows, will D still work? Explain your answer. (2)

5.2 Write down the energy conversion that takes place in: (a) a light bulb; (b) an electric motor. (3)

5.3 Why are electrical wires made of copper covered with plastic? (2)

5.4 Name TWO different types of switches. (2)

5.5 (a) What happens when two north poles of magnets are brought close together, and when a north pole and a south pole are brought close together? (b) Which TWO of these materials are attracted to a magnet: iron, aluminium, copper, steel? (3)

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TOTAL: 75

END OF QUESTION PAPER